ABSTRACT OF THE DISCLOSURE

A novel computer system architecture wherein all major internal and external data buses are implemented using massively parallel optical interconnects. The architecture replaces conventional electrical pathways—including the CPU-to-motherboard socket, the memory bus, and peripheral interconnects—with optical interfaces composed of high-density photon emitter and sensor arrays. This unified optical data fabric eliminates the bottlenecks associated with electrical signaling, such as the CPU I/O limit and the "memory wall." The resulting system architecture provides an aggregate data bandwidth in the petabit-per-second (Pbps) range, enabling unprecedented performance for data-intensive applications and fostering new design paradigms such as high-density 3D chip stacking and tightly coupled heterogeneous computing.